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Structural properties of polysaccharides from cultivated fruit bodies and mycelium of *Cordyceps militaris*

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Abstract

The structural properties of polysaccharides, respectively, obtained from the fermented mycelium and cultivated fruiting bodies of the *Cordyceps militaris* were investigated and compared in this paper. First, the crude polysaccharides were extracted from the mycelium and the fruiting bodies, respectively. The polysaccharides were successively purified by Sevag and chromatography on Sephadex G-100 column to produce two polysaccharides fractions termed CMPS-II and CBPS-II, respectively. The average molecular weights of CMPS-II and CBPS-II were 1.402×10^3 kDa and 1.273×10^3 kDa, respectively, and they were mainly composed of mannose, glucose and galactose in the mole ratios of 1:28.63:1.41 and 1:12.41:0.74, respectively, for CMPS-II and CBPS-II. Afterward, the structural features of CMPS-II and CBPS-II were investigated by a combination of chemical and instrumental analysis, such as FT-IR, periodate oxidation-Smith degradation, GC-MS, NMR and methylation analysis. The results indicated that structurally, both CMPS-II and CBPS-II were 1,3-branched-galactomannoglucan that had a linear backbone of (1→4)-linked α -D-glucopyranose (GlcP). Congo-red test revealed that CMPS-II and CBPS-II existed as triple-helical chains in 0.05–0.15 M NaOH solution.

Keywords: *Cordyceps militaris*; Polysaccharide; Structure.

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